



Let's build Knowledge Graph

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RAG

RAG

- Retrieve information from a data source
- Augment information to LLM/SLM/Generative AI Models
- Generate final response

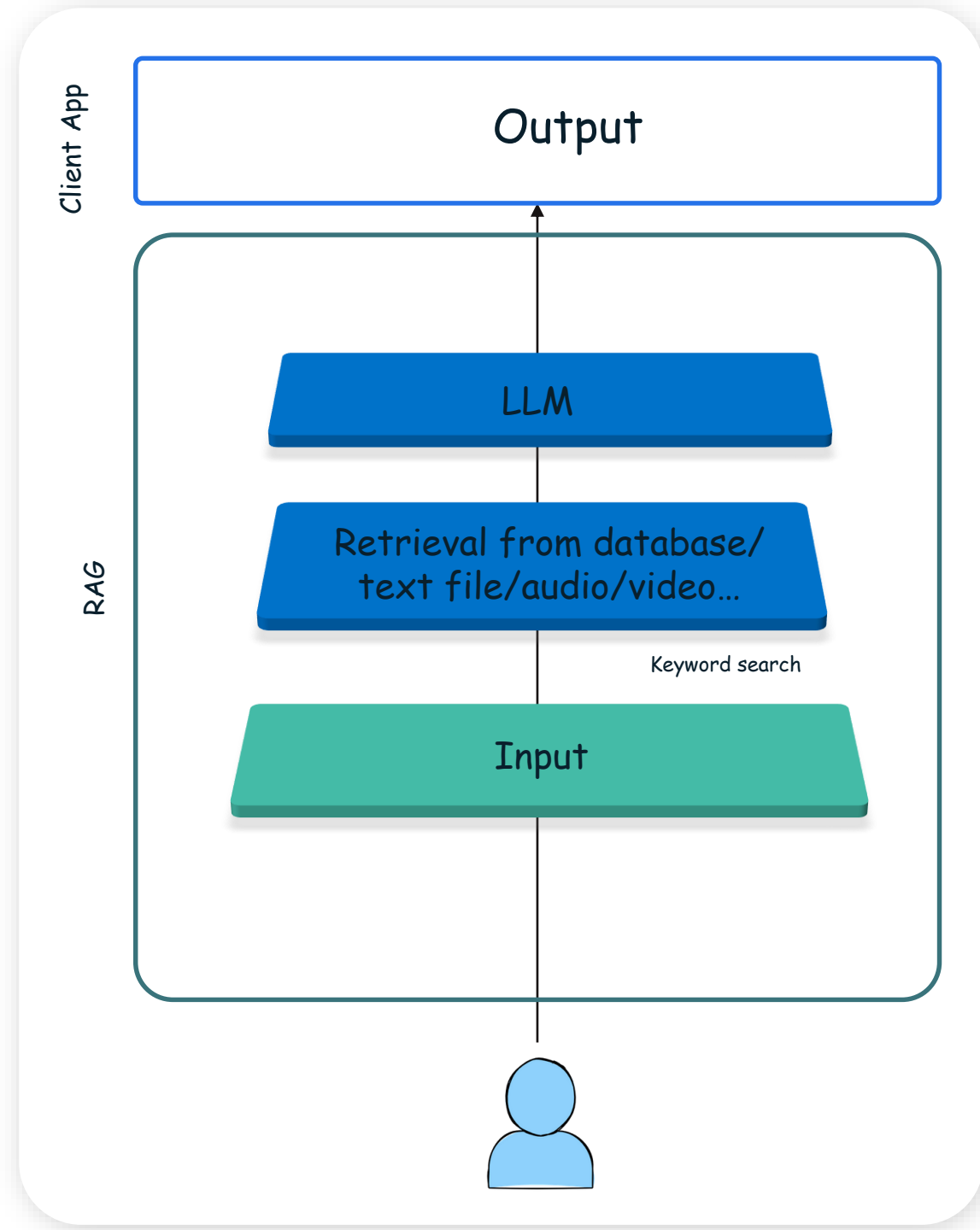
Retrieve

Augment

Generate

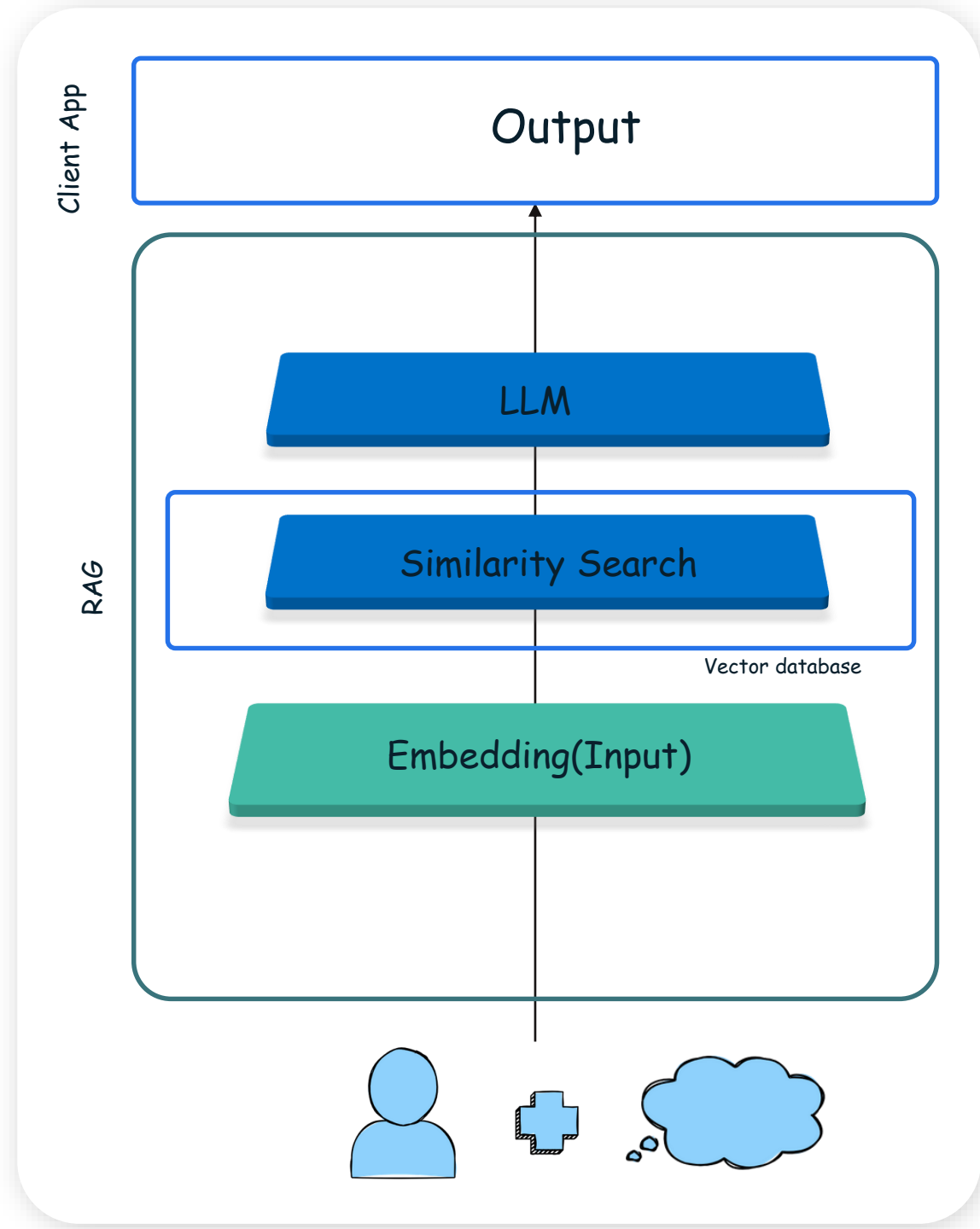
Naive RAG

- Provide input
- Retrieve information from a database
- Augment the additional context retrieved to LLM along with existing context
- Provide final output



Vector RAG

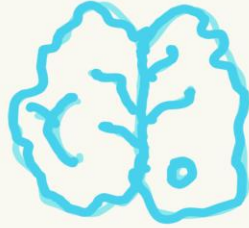
- Convert input query to embedding format
- Do vector similarity search on the database to retrieve relevant documents
- Provide context to LLM
- Generate final output



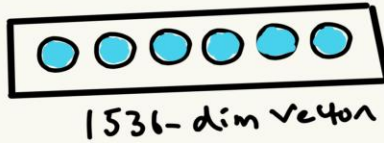
①
User Question



②
Embed Question

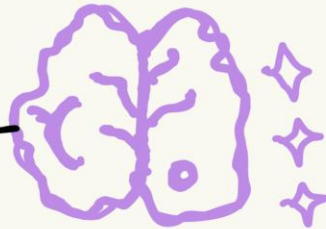


Question is converted into a 1536-dimension vector.

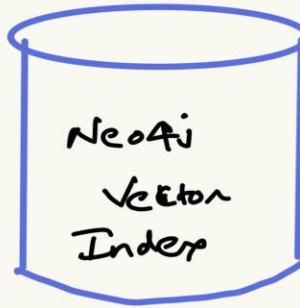


LLM uses retrieved text to generate final answer

⑥
Generate Answer

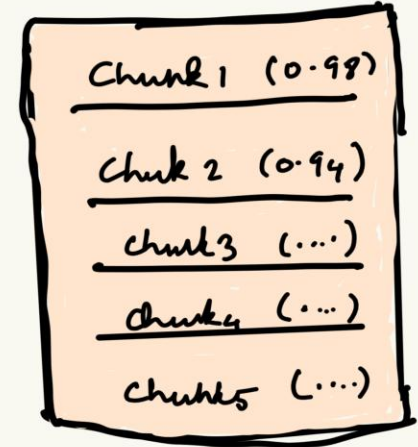


③
Vector Similarity Search



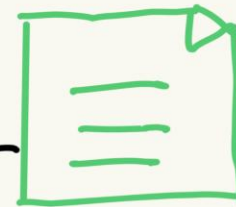
Vector is used to search to find the most similar chunks

④
Retrieve top chunks



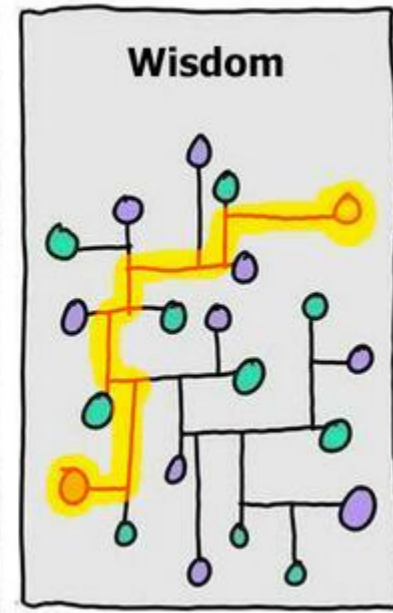
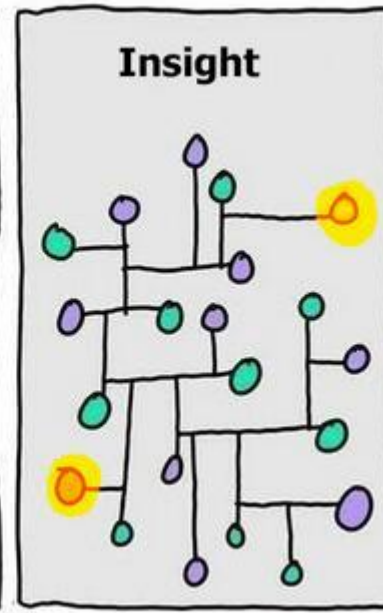
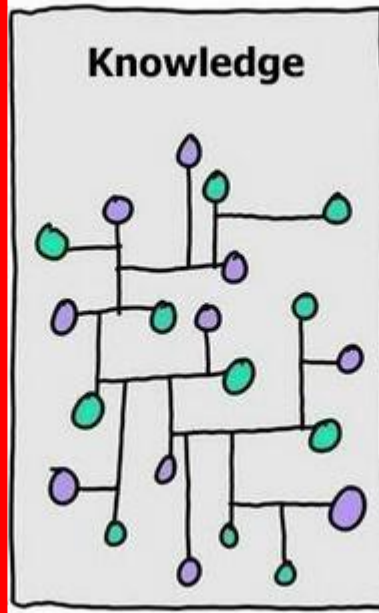
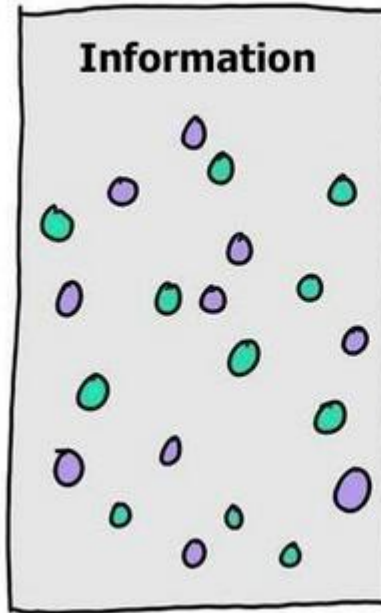
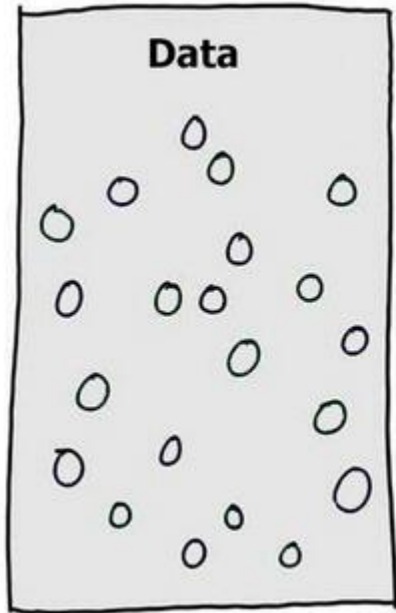
Top 5 most similar chunks are retrieved with scores

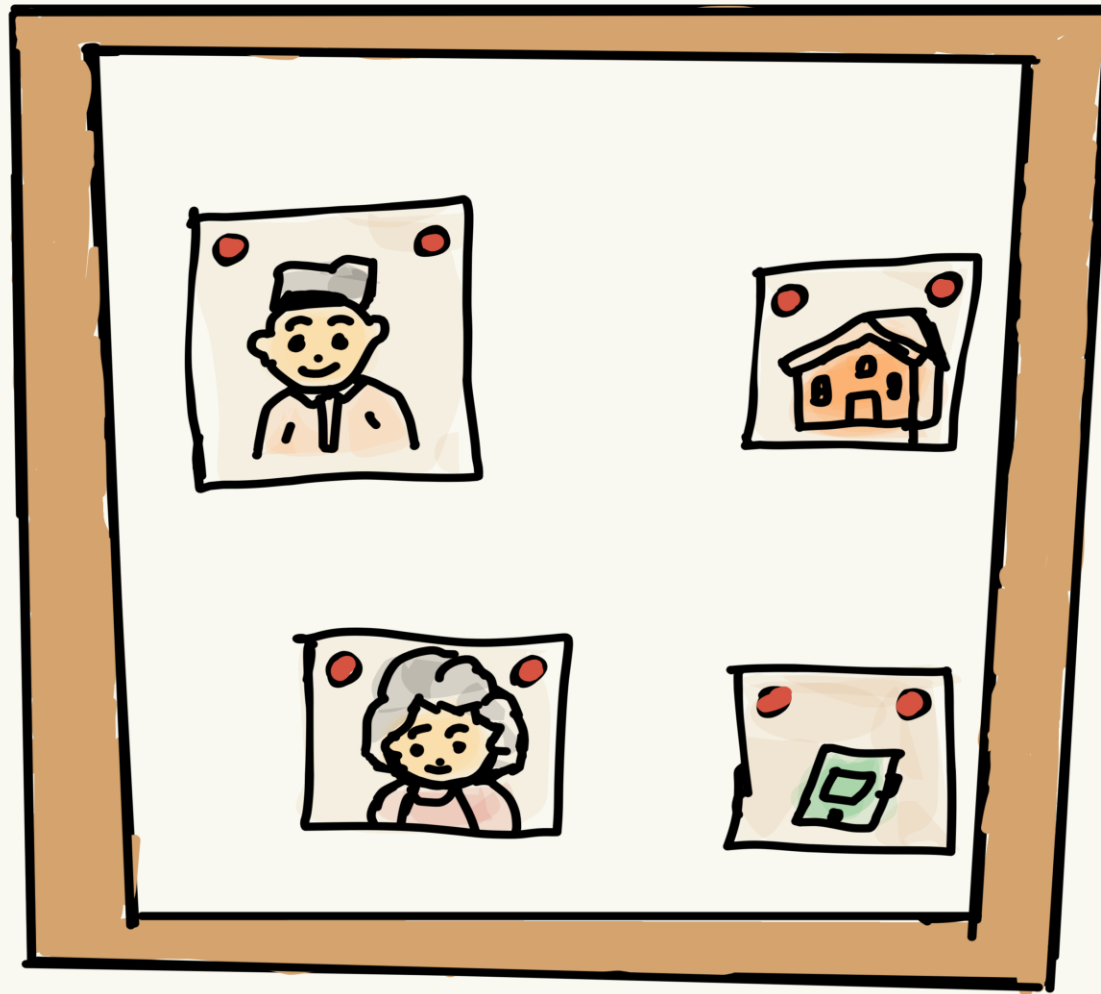
⑤
Extract Text



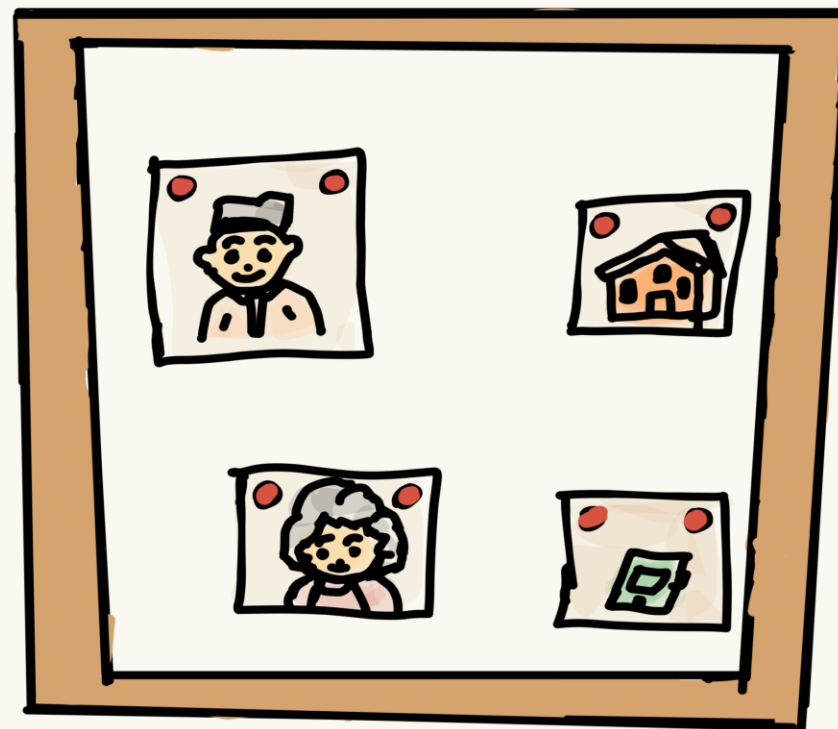
Text content is extracted from chunks

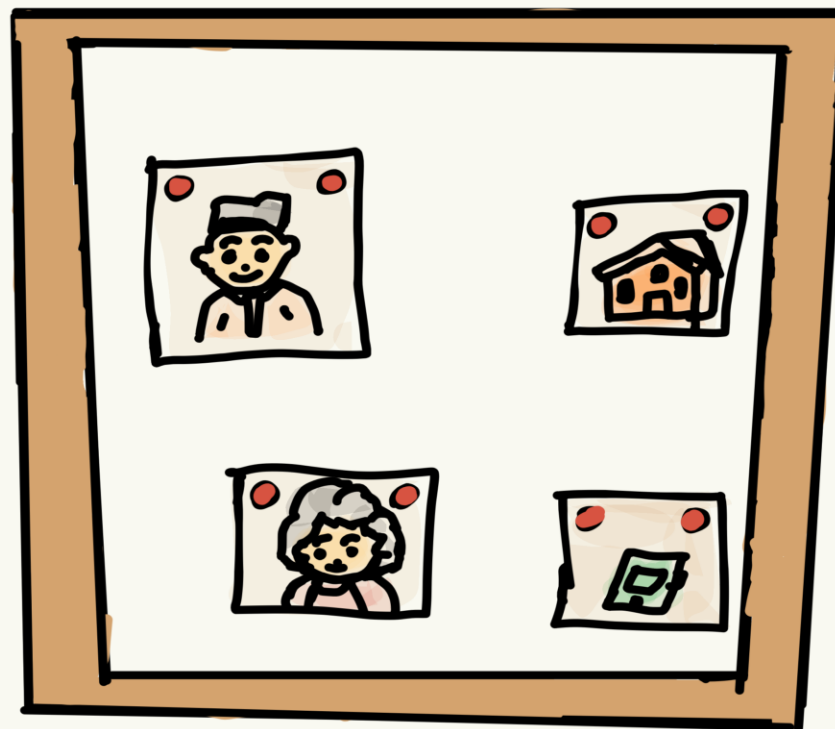




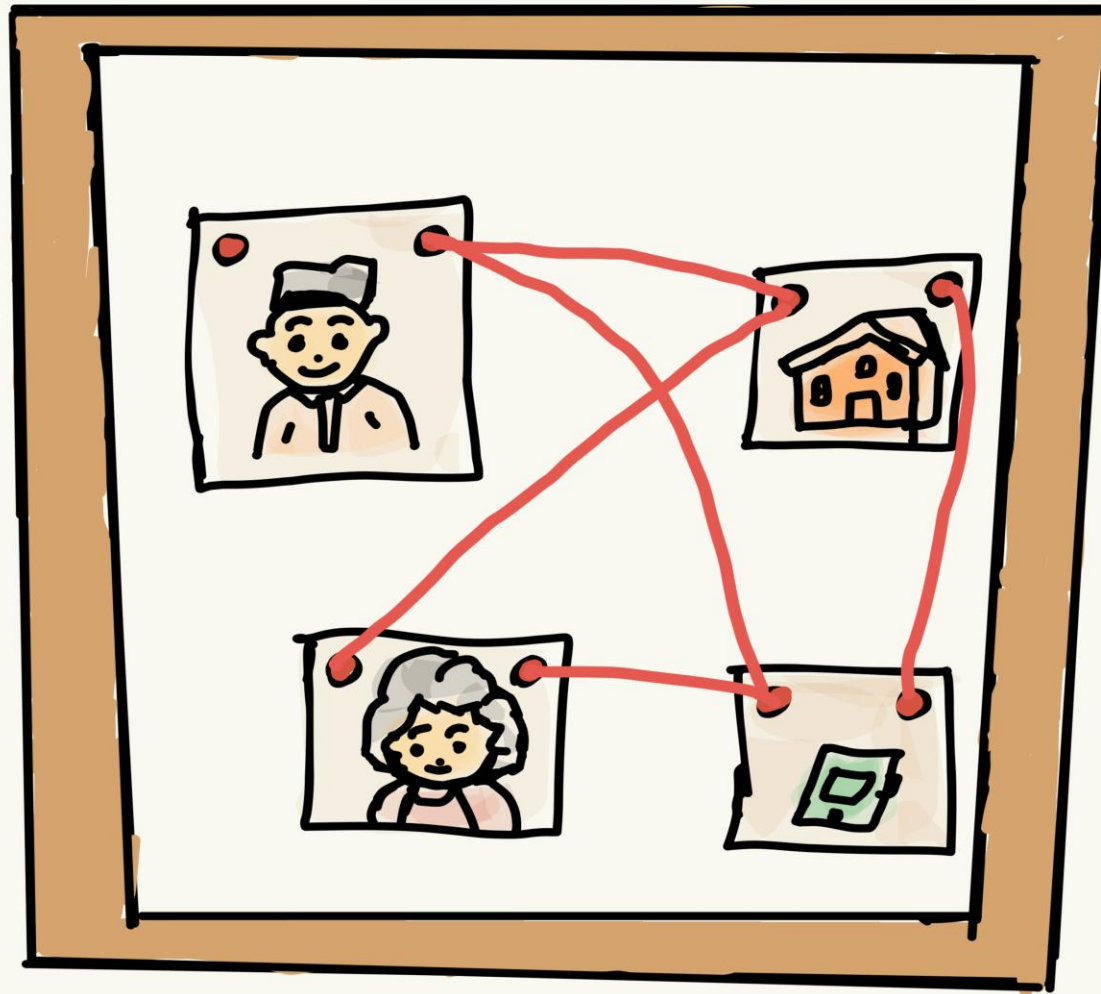


Giant Board

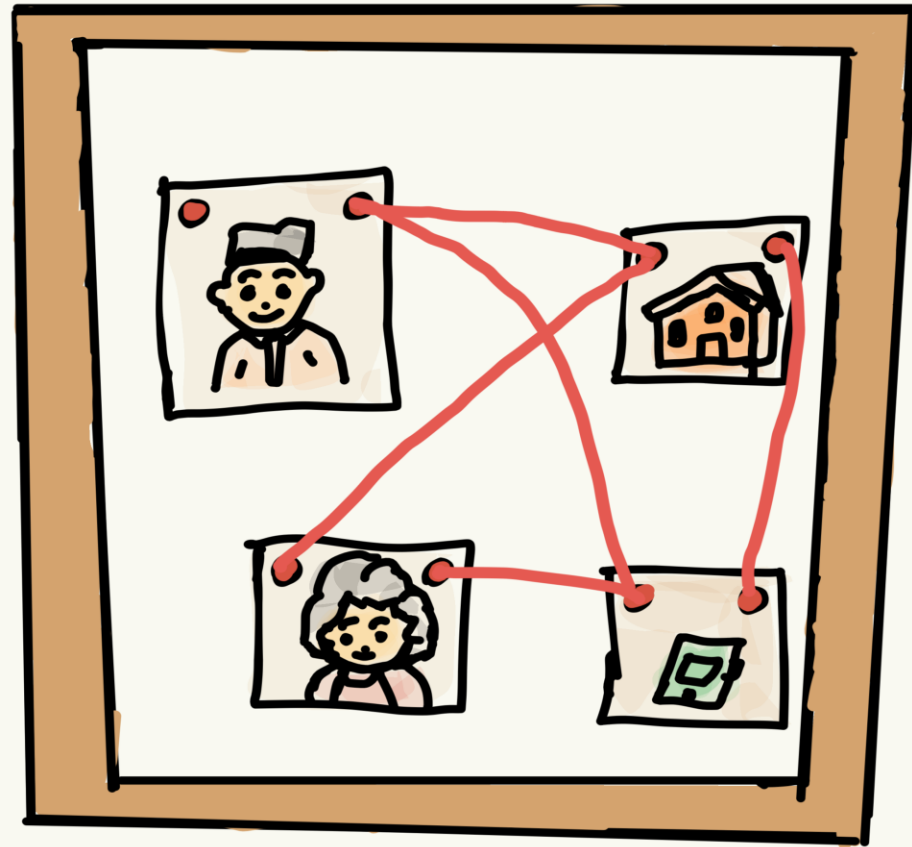




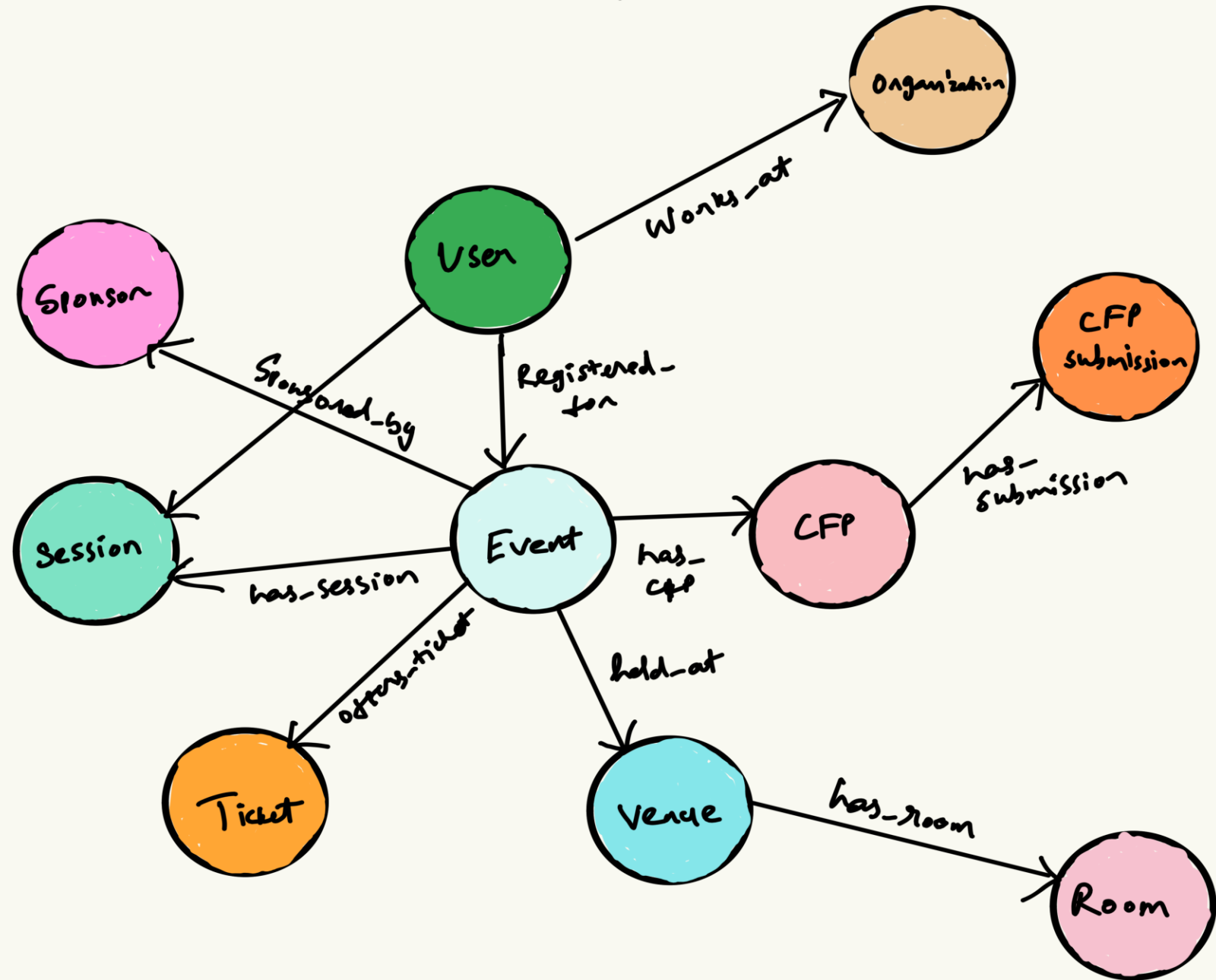
Something missing?

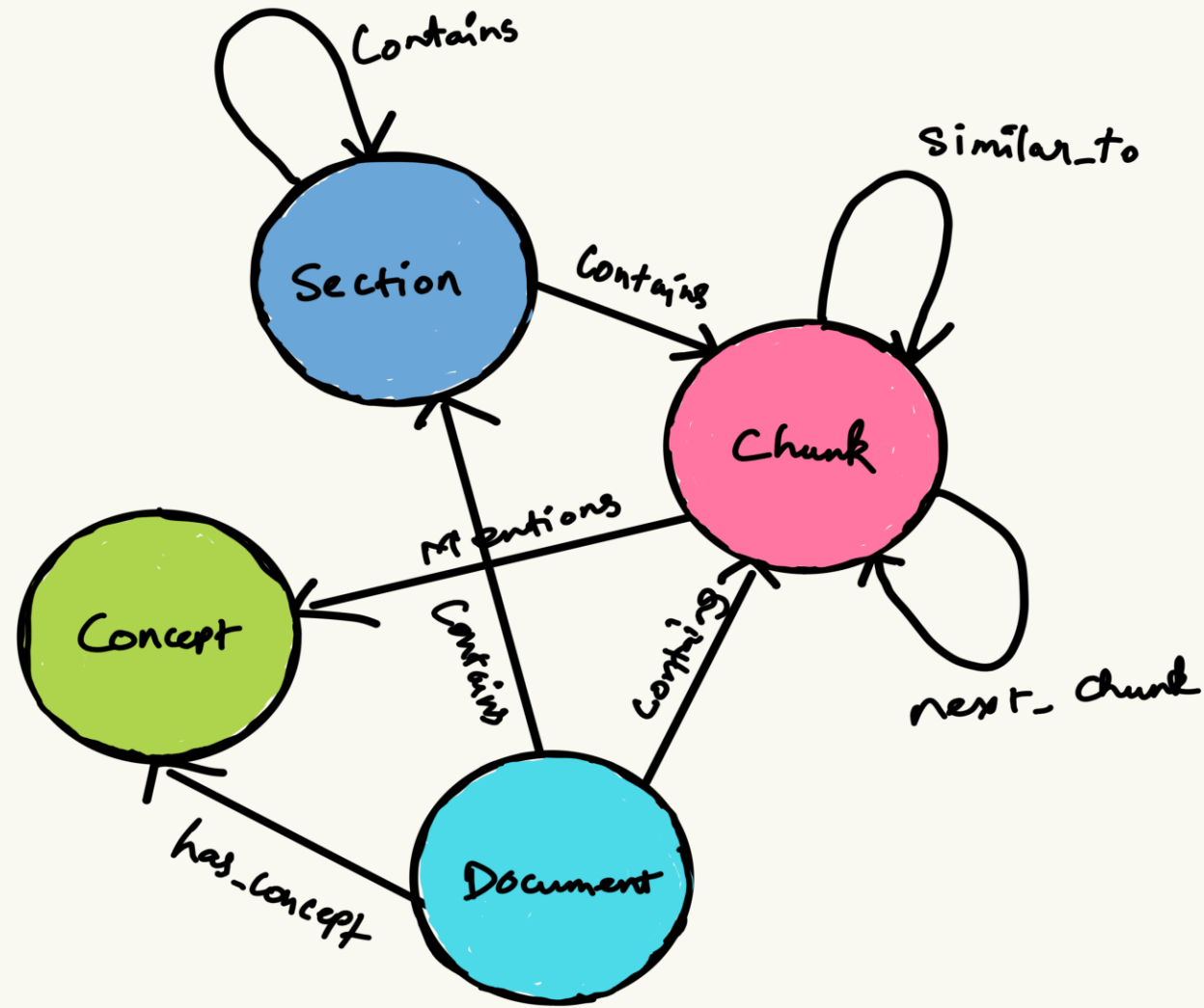


Strings



Domain Graph



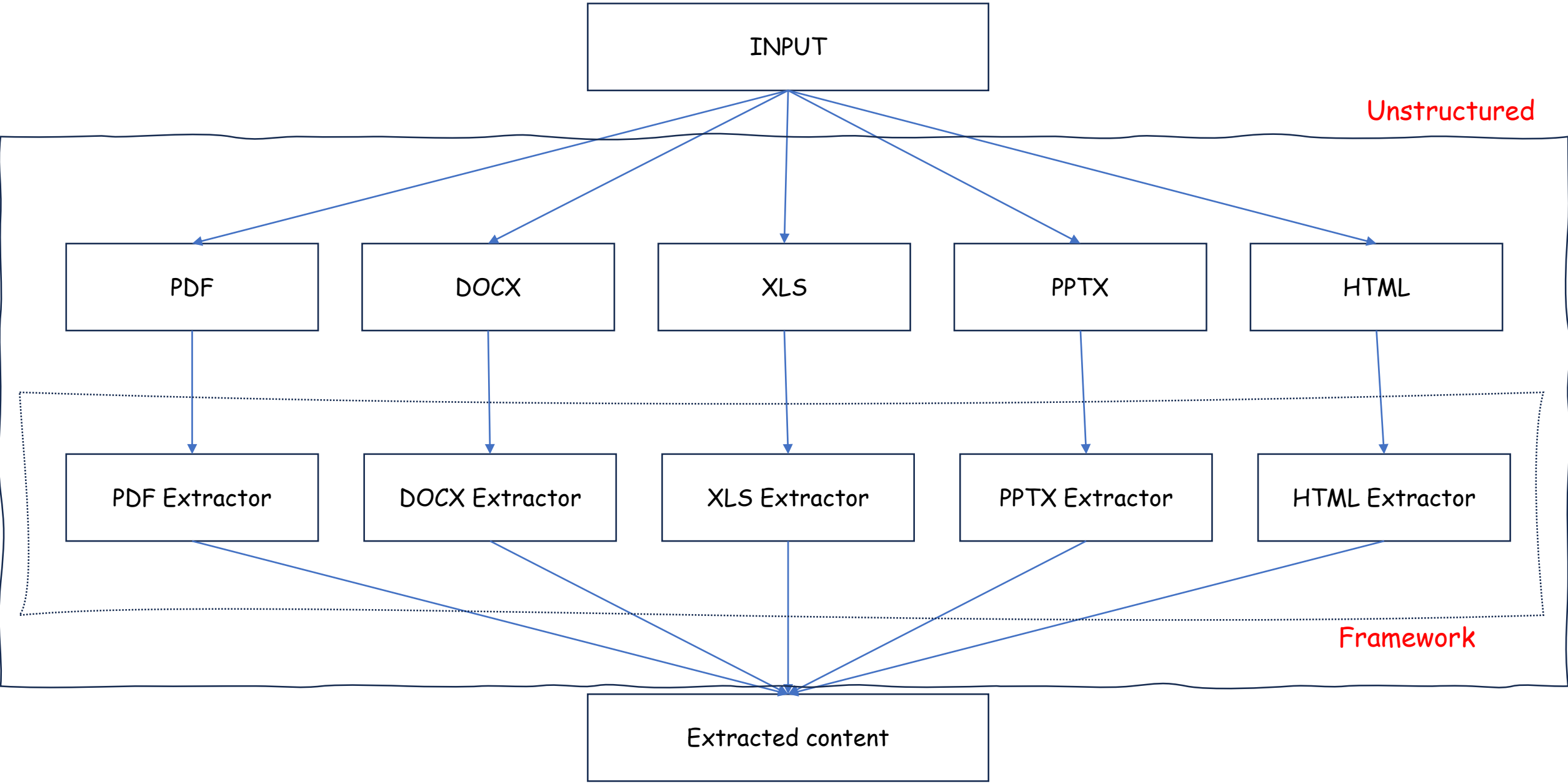


Lexical Graph

Ingestion

Ingestion Phase of building Lexical Graph

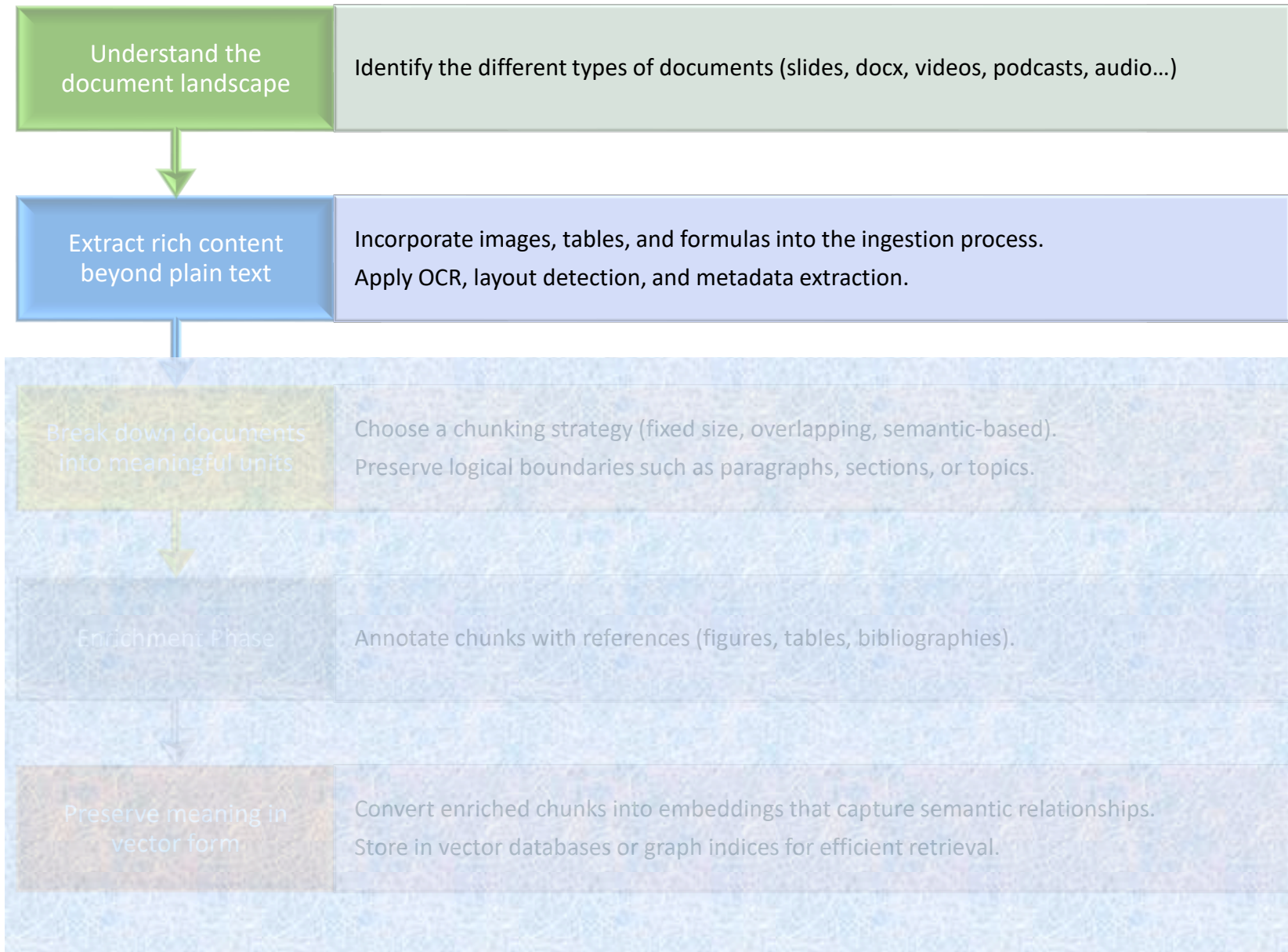


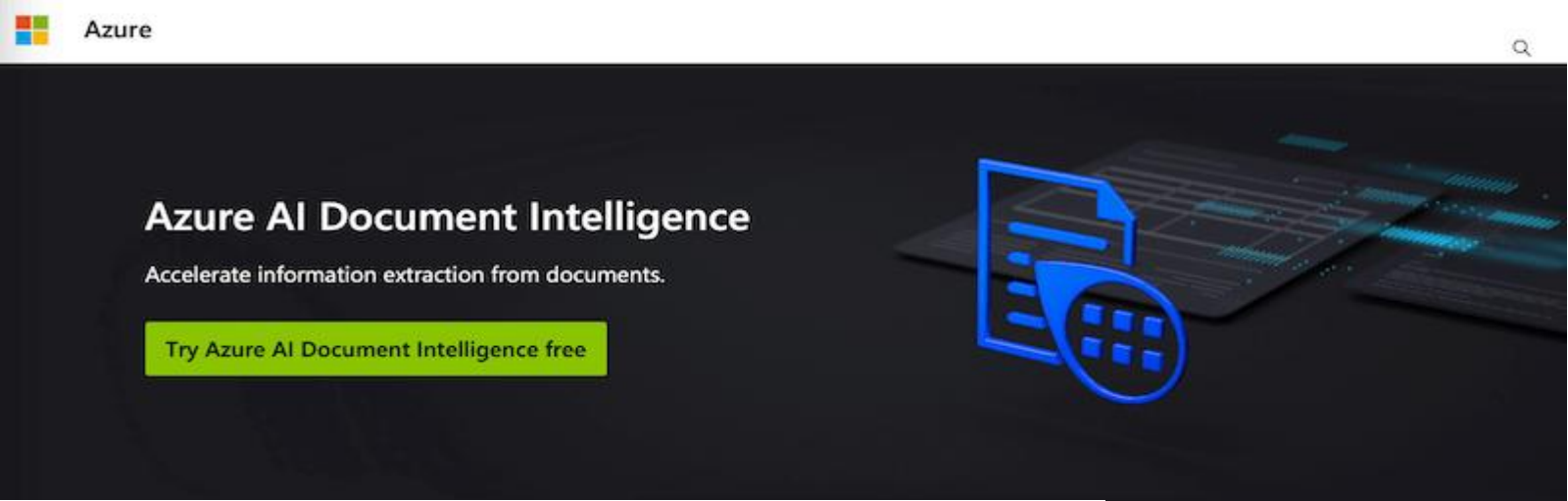




- Research on the supported libraries for proceeding with respective extractions
- Need to maintain custom code for the unsupported frameworks
- Often PDFs/DOCXs or any other unstructured documents doesn't only contain texts
- Research on how to extract images/tables/formulas

Ingestion Phase of building Lexical Graph





Azure AI Document Intelligence

Accelerate information extraction from documents.

Try Azure AI Document Intelligence free



0 This is title

1 1. Text
Latin refers to an ancient Italic language originating in the region of Latium in ancient Rome.

2 2. Page Objects
Here's a sample table below, designed to be simple for easy understand and quick reference.

3 2.1 Table

Name	Corp	Remark
Foo		
Bar	Microsoft	Dummy

Table 1: This is a dummy table

4 2.2. Figure

Figure 1: Here is a figure with text

5 3. Others
AI Document Intelligence is an AI service that applies advanced machine learning to extract text, key-value pairs, tables, and structures from documents automatically and accurately:

- ☒ clear
- ☒ precise
- ☐ vague
- ☒ coherent
- ☐ incomprehensible

Turn documents into usable data and shift your focus to acting on information rather than compiling it. Start with prebuilt models or create custom models tailored to your documents both on premises and in the cloud with the AI Document Intelligence studio or SDK.

Learn how to accelerate your business processes by automating text extraction with AI Document Intelligence. This webinar features hands-on demos for key use cases such as document processing, knowledge mining, and industry-specific AI model customization.

```
"sections": [
  0 {
    "sections/0"
    "spans": [ ... ],
    "elements": [
      1 "/paragraphs/1",
      2 "/sections/2",
      5 "/sections/5"
    ]
  },
  "spans": [ ... ],
  "elements": [ ... ],
  "spans": [ ... ],
  "elements": [ ... ],
  "spans": [ ... ],
  "elements": [
    3 "/paragraphs/4",
    4 "/sections/3",
    4 "/sections/4"
  ]
]
```

Page header → This is the header of the c

Title → This is title

Section heading → 1. Text
Latin refers to an ancient Italic language originating in the region of Latium in ancient Rome.

Paragraph → 2. Page Objects
Here's a sample table below, designed to be simple for easy understand and quick reference.

Subsection heading → 2.1 Table

Selection marks →

Name	Corp	Remark
Foo		
Bar	Microsoft	Dummy

Table → Table 1: This is a dummy table

Table caption → 2.2. Figure

Figure caption → Figure 1: Here is a figure with text

Figure →

Page number → 1 | Page

Page footer → This is the footer of the document.

3. Others
AI Document Intelligence is an AI service that applies advanced machine learning to extract text, key-value pairs, tables, and structures from documents automatically and accurately:

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- ☐ vague
- ☒ coherent
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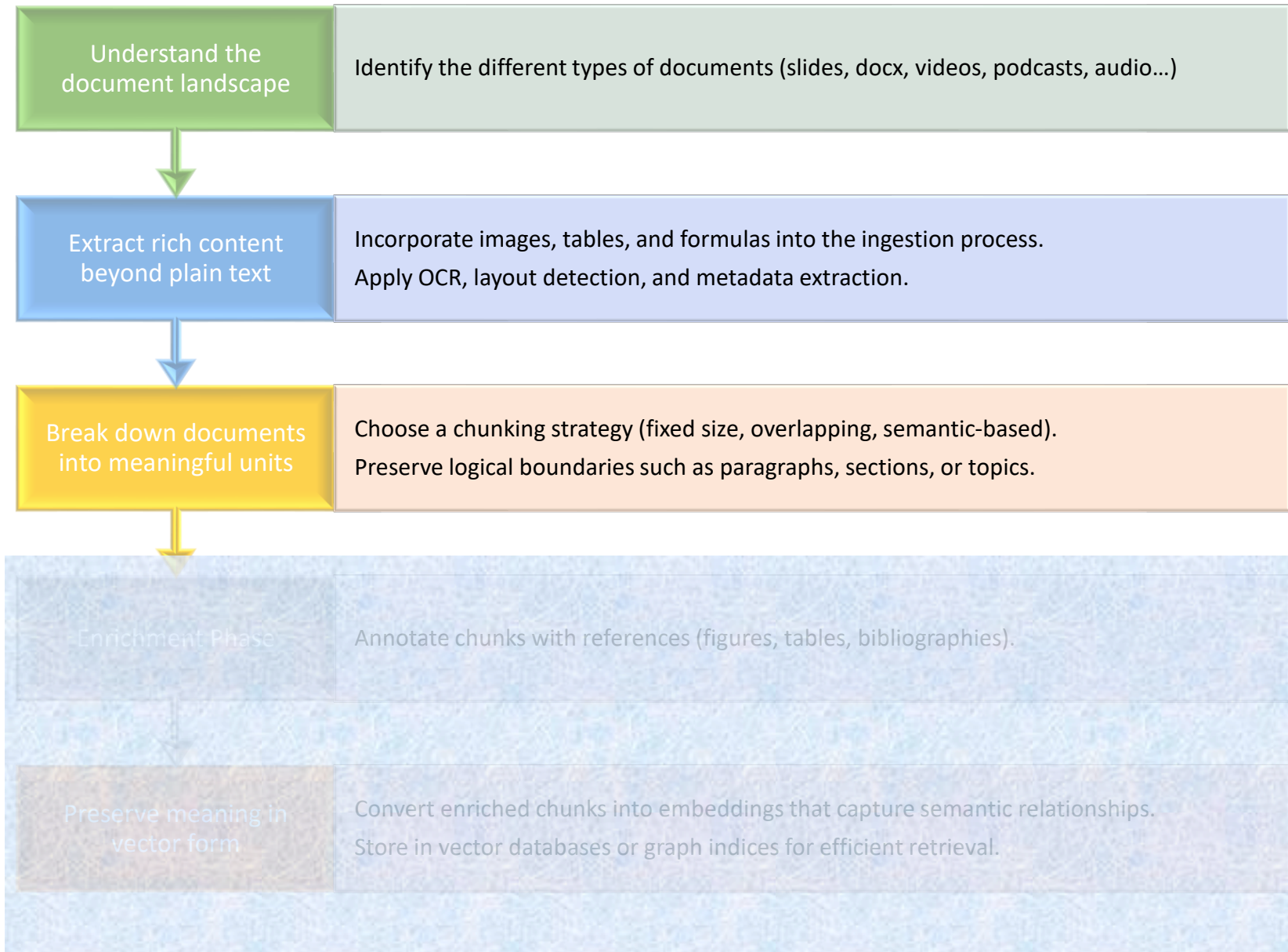
2.2. Figure
1 Figure 1: Here is a figure with text
2
3
4

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      1 {
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          3.591,
          6.9657,
          1.4183,
          6.9657
        ]
      },
      2 {
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          4.1763,
          7.1074,
          4.1781,
          9.0873,
          1.0324,
          9.0891
        ]
      }
    ]
  }
]
```



- Verbalize image
- Transform table rows, cells, columns to a markdown representation
- Verbalize formulas
- Track the pages where it is referenced, if possible capture the captions of them

Ingestion Phase of building Lexical Graph



Chunking

Fixed Chunking, w/o overlap

- Easier to break down a document based on this strategy
- Context missed / scattered

LLM based Chunking

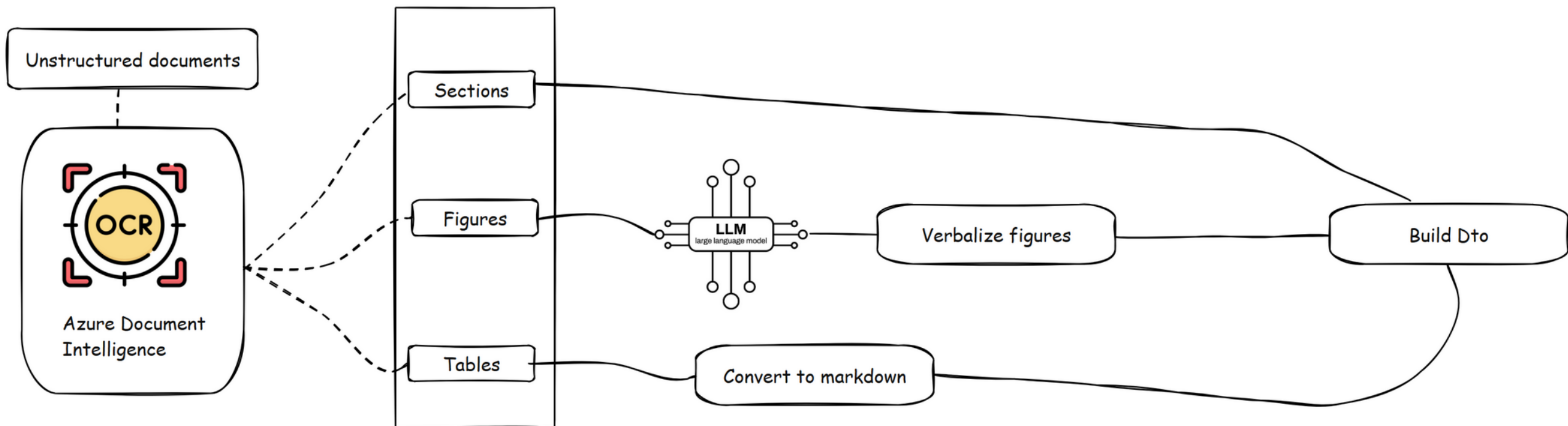
- Preserves the context
- Expensive

Semantic Chunking

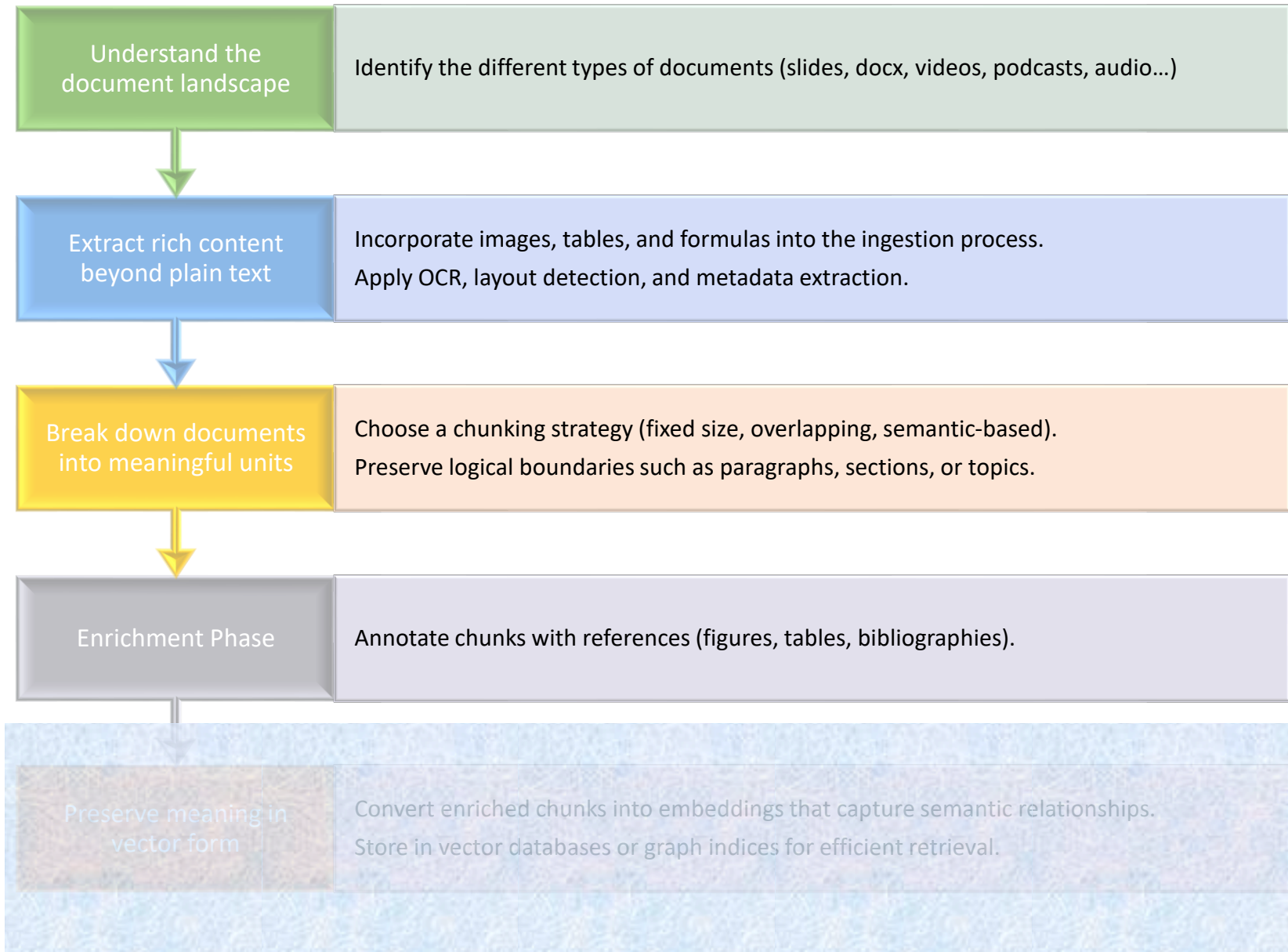
- Split document based on sentences, paragraphs, pages or sections
- Preserves the context
- Variable Chunk size

Magic Formula

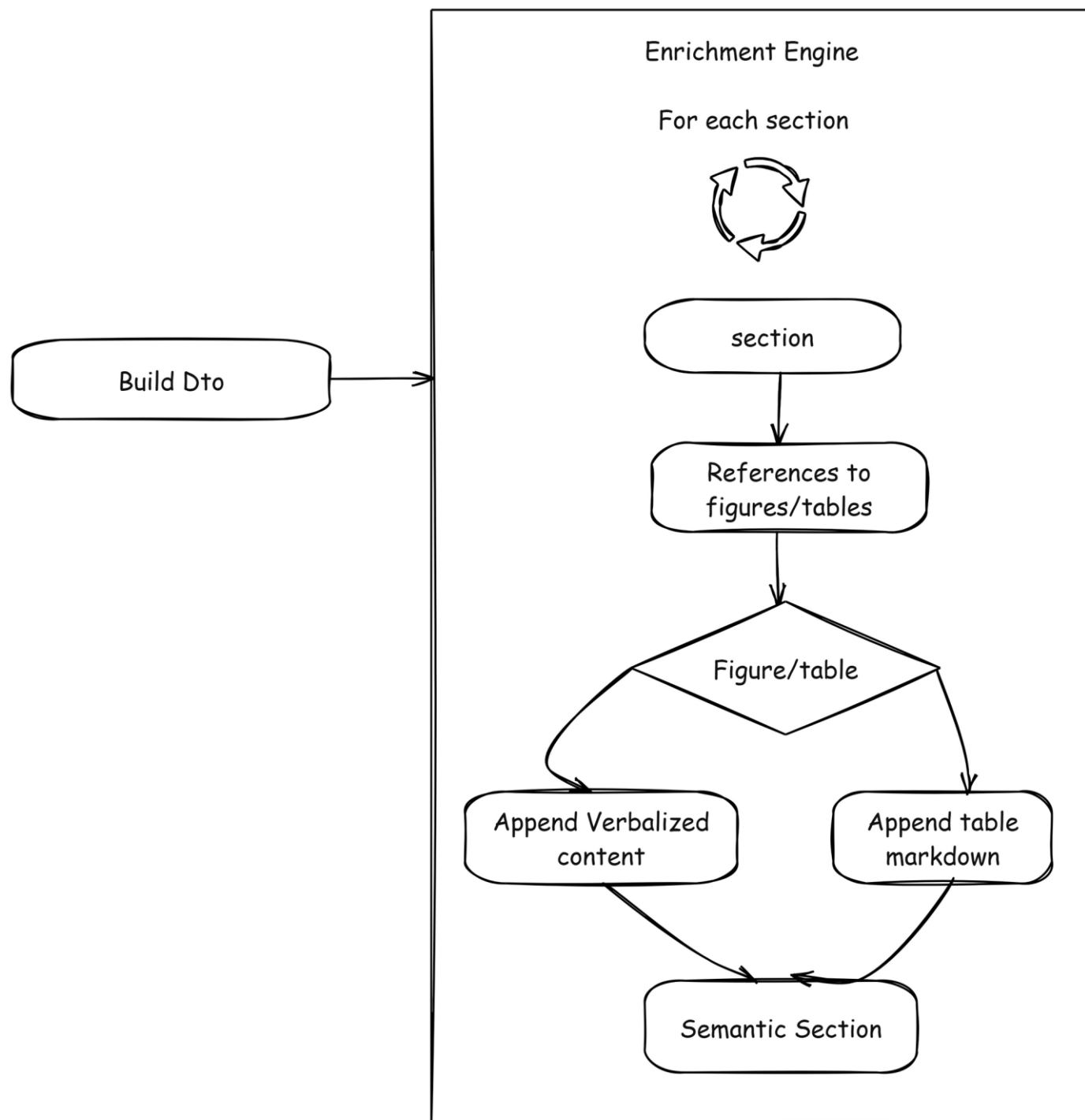




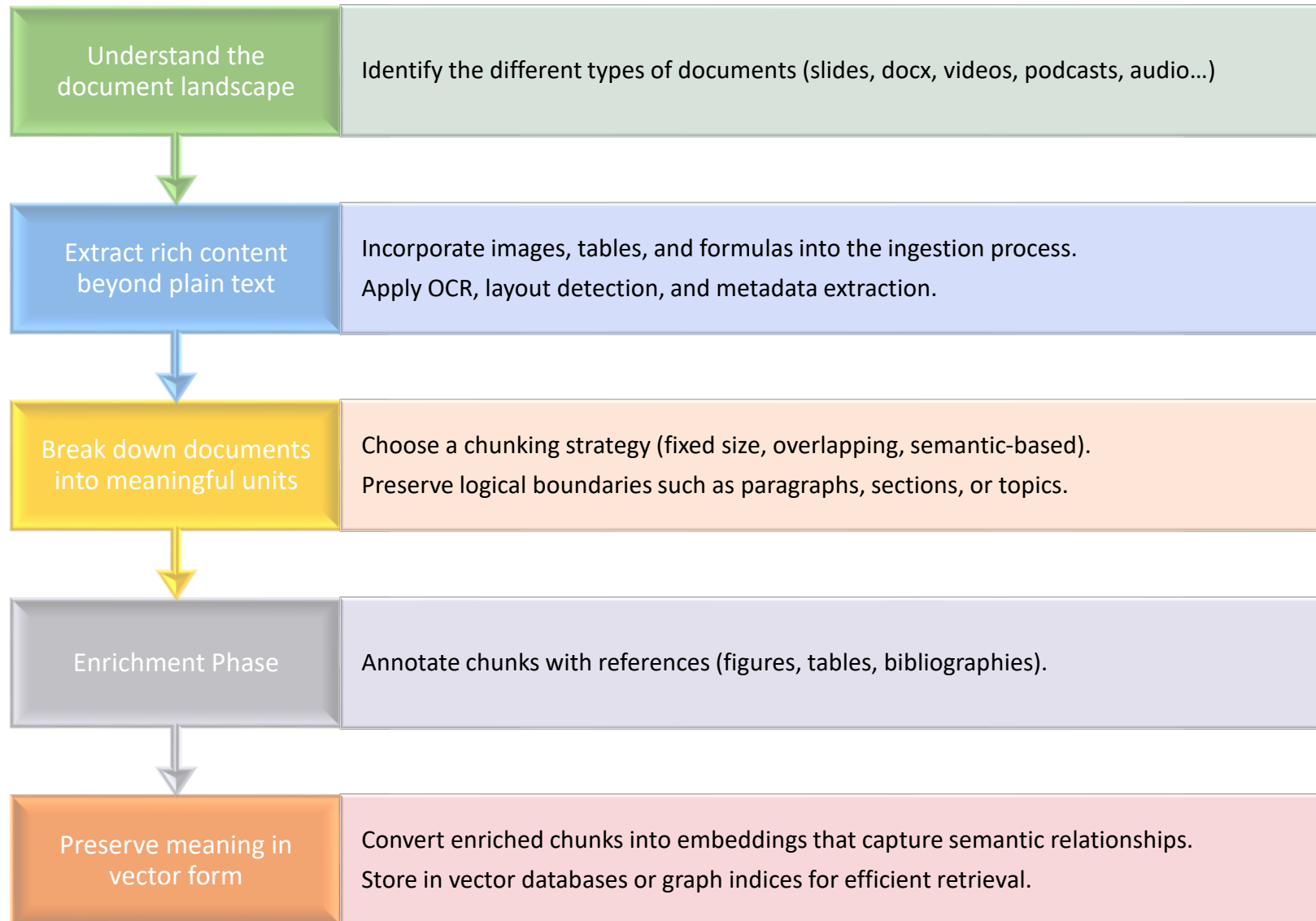
Ingestion Phase of building Lexical Graph



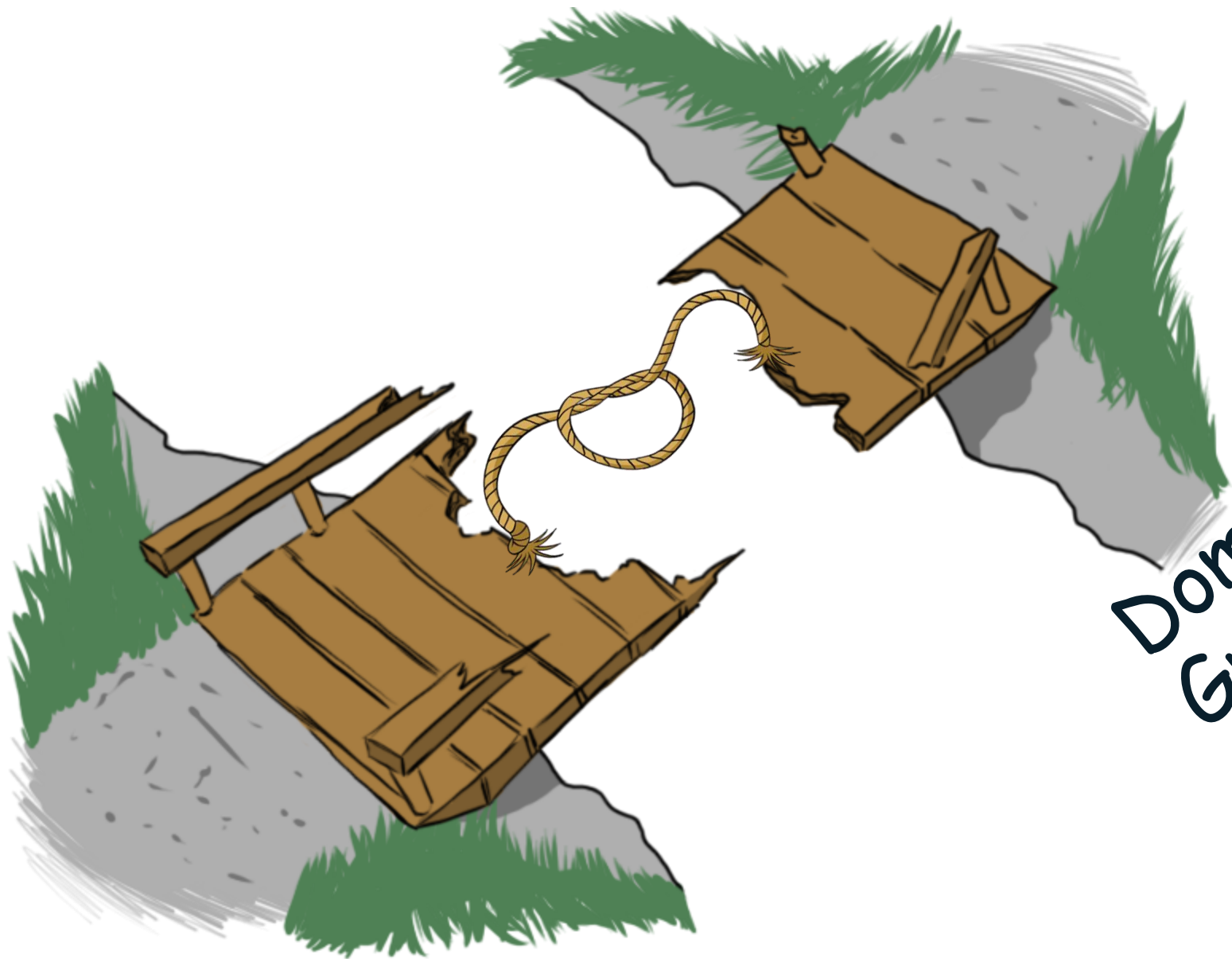
Enrichment



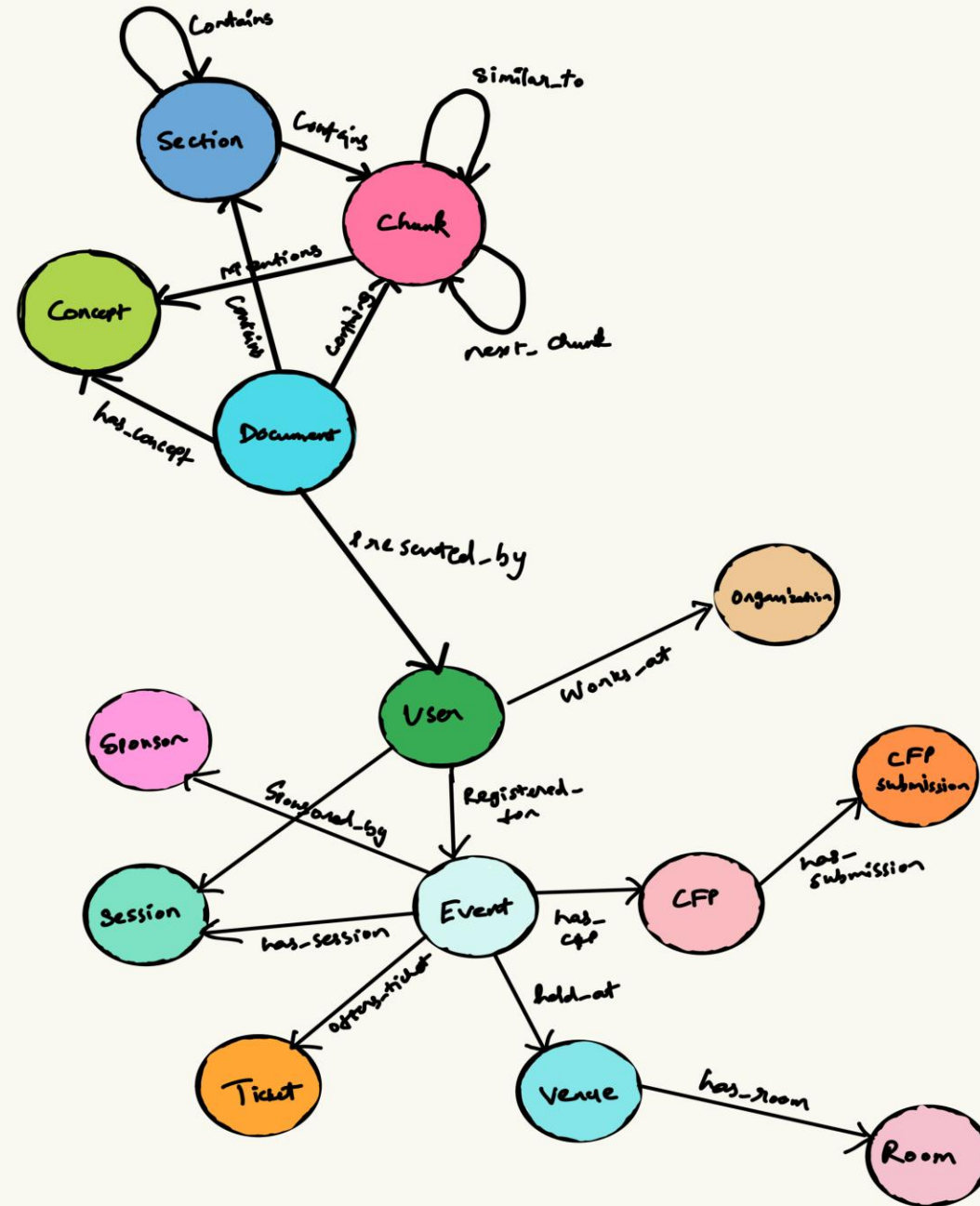
Ingestion Phase of building Lexical Graph

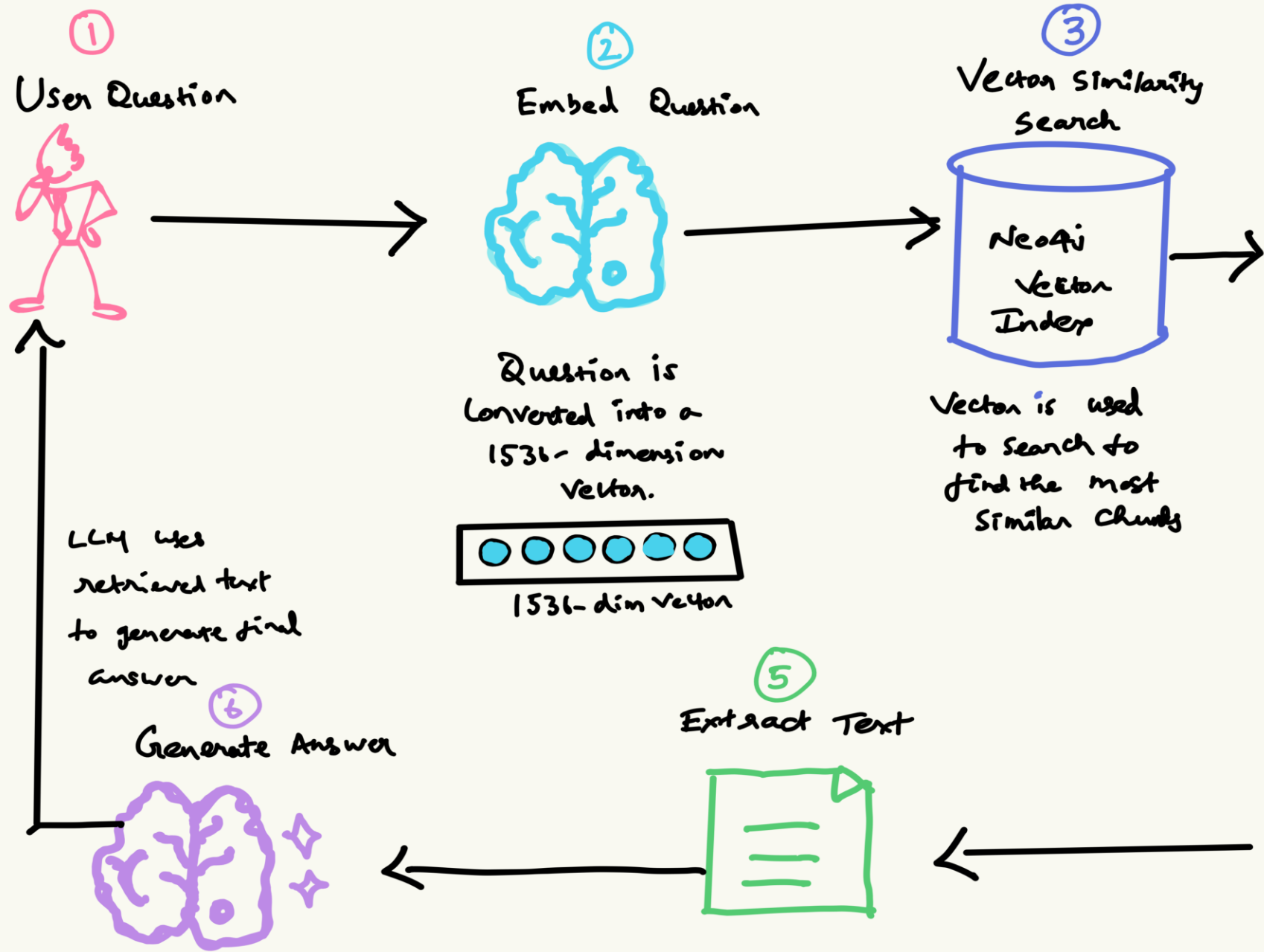


Lexical
Graph



Domain
Graph



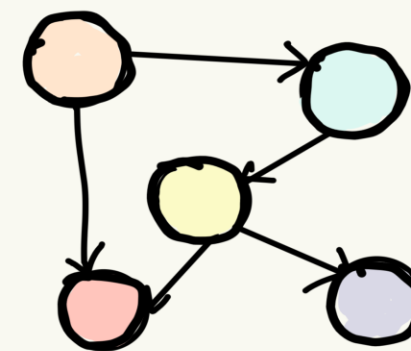


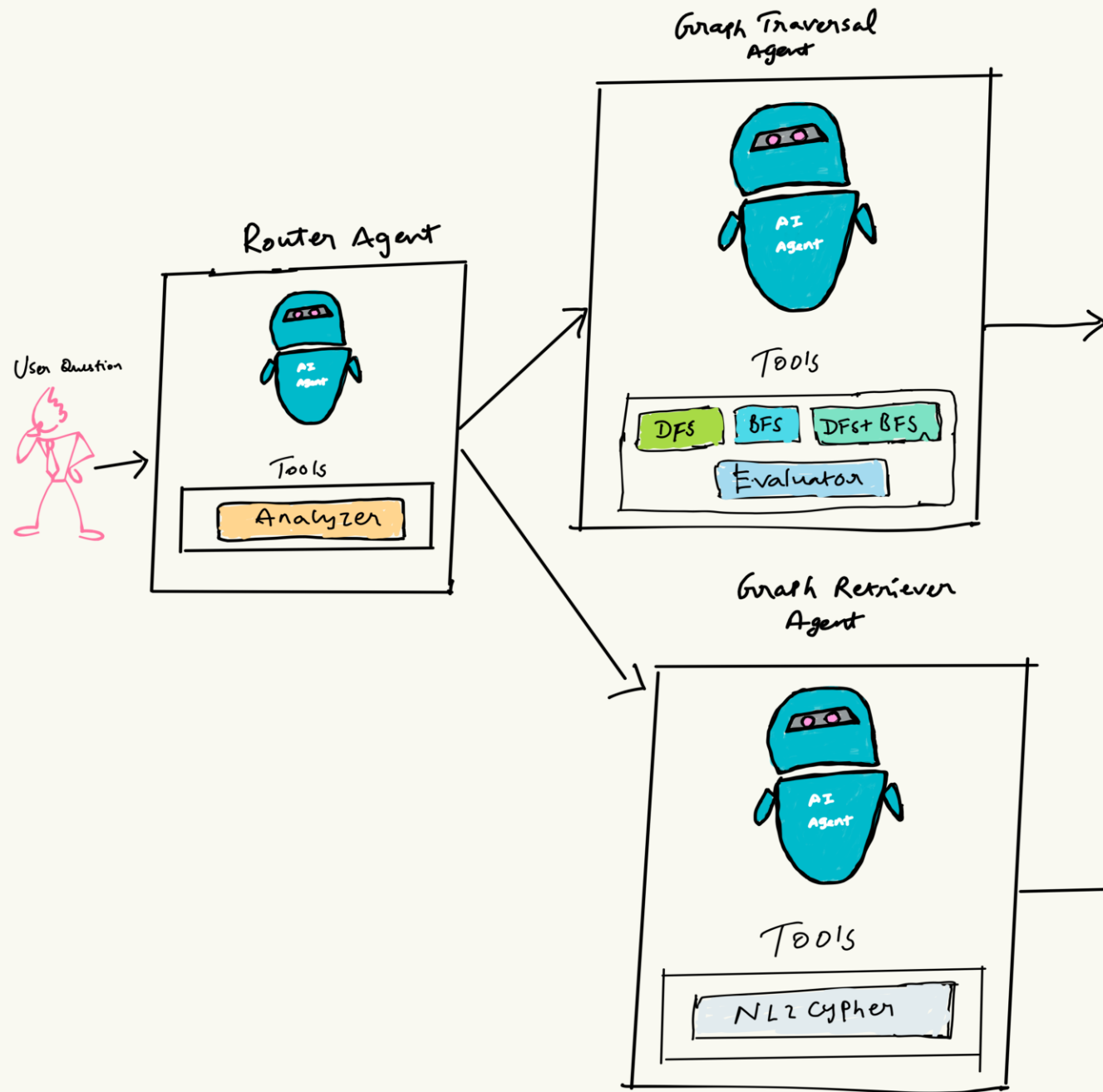
4.1 Retrieve top chunks

Chunk 1	(0.99)
Chunk 2	(0.94)
Chunk 3	(...)
Chunk 4	(...)
Chunk 5	(...)

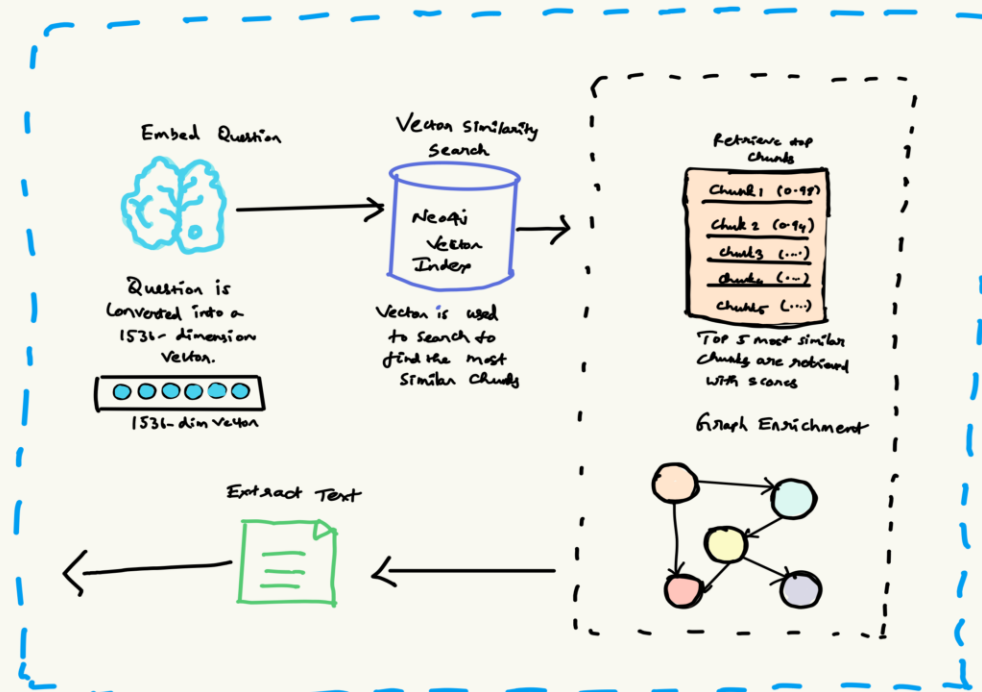
Top 5 most similar chunks are retrieved with scores

4.2 Graph Enrichment

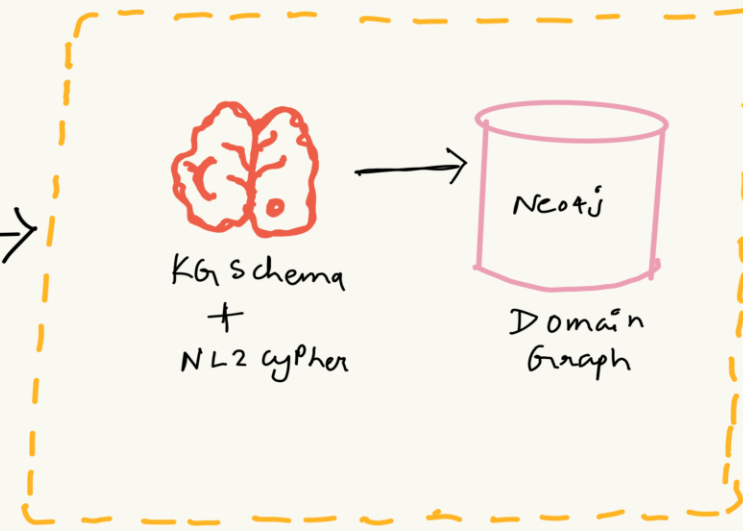




DFS/BFS Retrieval Path



NL2 Cypher Path



Thank You



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